

Lakshya JEE 2027

PYQ Practice Sheet

Physics

Electrostatics

- Q1** Sixty four conducting drops each of radius 0.02 m and each carrying a charge of $5 \mu\text{C}$ are combined to form a bigger drop. The ratio of surface charge density of bigger drop to the smaller drop will be:

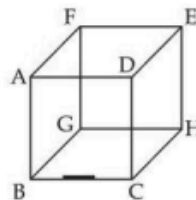
(A) 1 : 4 (B) 4 : 1
(C) 1 : 8 (D) 8 : 1

- Q2** An electron is made to enter symmetrically between two parallel and equally but oppositely charged metal plates, each of 10 cm length. The electron emerges out of the electric field region with a horizontal component of velocity 10^6 m/s . If the magnitude of the electric field between the plates is 9.1 V/cm , then the vertical component of velocity of electron is: (mass of electron = $9.1 \times 10^{-31} \text{ kg}$ and charge of electron = $1.6 \times 10^{-19} \text{ C}$)

[JEE Main 2025, 22 Jan (Shift 1)]

(A) $16 \times 10^6 \text{ m/s}$
(B) 0 m/s
(C) $16 \times 10^4 \text{ m/s}$
(D) $1 \times 10^6 \text{ m/s}$

- Q3** A line charge of length $\frac{a}{2}$ is kept at the center of an edge BC of a cube $ABCDEFGH$ having edge length 'a' as shown in the figure. If the density of line charge is $\lambda \text{ C}$ per unit length, then the total electric flux through all the faces of the cube will be ____ - (Take, ϵ_0 as the free space permittivity)



[JEE Main 2025, 22 Jan (Shift 1)]

(A) $\frac{\lambda a}{16\epsilon_0}$ (B) $\frac{\lambda a}{2\epsilon_0}$
(C) $\frac{\lambda a}{4\epsilon_0}$ (D) $\frac{\lambda a}{8\epsilon_0}$

- Q4** A dipole comprises of two charged particles of identical magnitude q and opposite in nature. The mass m of the positive charged particle is half of the mass of the negative charged particle. The two charges are separated by a distance l . If the dipole is placed in a uniform electric field $\vec{E}$; such a way that dipole axis makes a very small angle with the electric field $\vec{E}$. The angular frequency of the oscillations of the dipole when released is given by:

[JEE Main 2023, 6 Apr (Shift 2)]

(A) $\sqrt{\frac{8qE}{3ml}}$
(B) $\sqrt{\frac{4qE}{ml}}$
(C) $\sqrt{\frac{4qE}{3ml}}$
(D) $\sqrt{\frac{3qE}{2ml}}$

- Q5** A $10 \mu\text{C}$ charge is divided into two parts and placed at 1 cm distance so that the repulsive force between them is maximum. The charges of the two parts are:

[JEE Main 2023, 13 Apr (Shift 2)]

(A) $9 \mu\text{C}$, $1 \mu\text{C}$ (B) $5 \mu\text{C}$, $5 \mu\text{C}$
(C) $7 \mu\text{C}$, $3 \mu\text{C}$ (D) $9 \mu\text{C}$, $2 \mu\text{C}$

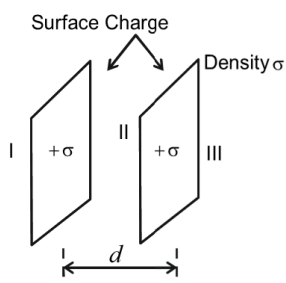
Q6



Charge is distributed within a sphere of radius R with a volume charge density $\rho(r) = \frac{A}{r^2} e^{-2r/a}$, where A and a are constants. If Q is the total charge of this charge distribution the radius R is:

- (A) $a \log \left(1 - \frac{Q}{2\pi a A} \right)$
 (B) $\frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$
 (C) $\frac{a}{4} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$
 (D) $\frac{a}{2} \log \left(1 - \frac{1}{2\pi a A} \right)$

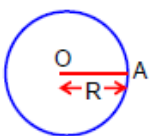
- Q7** Let σ be the uniform surface charge density of two infinite thin plane sheets shown in figure. Then the electric fields in the three different region E_I , E_{II} and E_{III} are.



[JEE Main 2023, 1 Feb (Shift 1)]

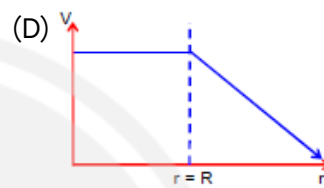
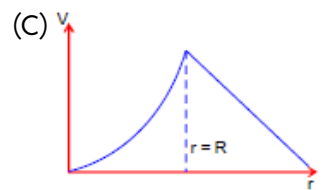
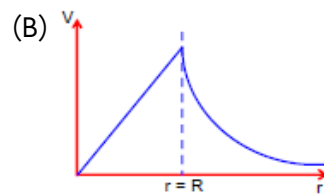
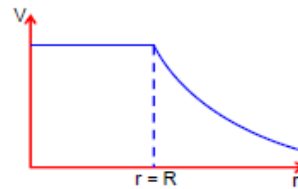
- (A) $\vec{E}_I = -\frac{\sigma}{2\epsilon_0} \hat{n}$, $\vec{E}_{II} = 0$, $\vec{E}_{III} = \frac{\sigma}{2\epsilon_0} \hat{n}$
 (B) $\vec{E}_I = 0$, $\vec{E}_{II} = \frac{\sigma}{\epsilon_0} \hat{n}$, $\vec{E}_{III} = 0$
 (C) $\vec{E}_I = \frac{2\sigma}{\epsilon_0} \hat{n}$, $\vec{E}_{II} = 0$, $\vec{E}_{III} = \frac{2\sigma}{\epsilon_0} \hat{n}$
 (D) $\vec{E}_I = -\frac{\sigma}{\epsilon_0} \hat{n}$, $\vec{E}_{II} = 0$, $\vec{E}_{III} = \frac{\sigma}{\epsilon_0} \hat{n}$

- Q8** For a uniformly charged thin spherical shell, the electric potential (V) radially away from the centre (O) of shell can be graphically represented as -



[JEE Main 2023, 6 Apr (Shift 1)]

(A)



- Q9** An electric dipole is placed on x -axis in proximity to a line charge of linear density $3.0 \times 10^{-6} \text{ C/m}$. Line charge is placed on z -axis and positive and negative charge of dipole is at a distance of 10 mm and 12 mm from the origin respectively. If total force of 4 N is exerted on the dipole, find out the amount of positive charge of the dipole.

[JEE Main 2021, 22 Jul (Shift 2)]

- (A) $0.485 \mu\text{C}$
 (B) $8.8 \mu\text{C}$
 (C) $815.1 \mu\text{C}$
 (D) $4.44 \mu\text{C}$

- Q10** The electric flux is $\phi = \alpha\sigma + \beta\lambda$ where λ and σ are linear and surface charge density, respectively. $\left(\frac{\alpha}{\beta}\right)$ represents

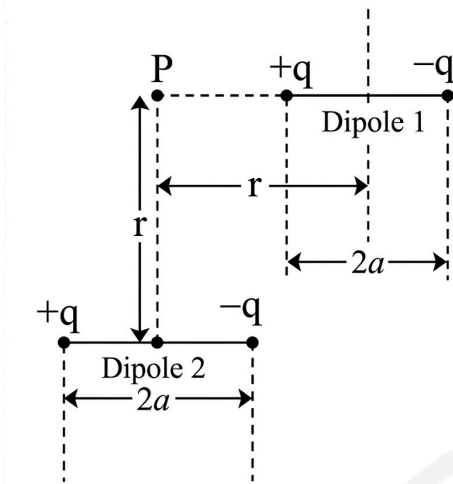
[JEE Main 2025, 23 Jan (Shift 1)]

- (A) charge (B) displacement
 (C) area (D) electric field

- Q11** A point particle of charge Q is located at P along the axis of an electric dipole 1 at a distance r as shown in the figure. The point P is also on the equatorial plane of a second electric dipole 2 at a



distance r . The dipoles are made of opposite charge q separated by a distance $2a$. For the charge particle at P not to experience any net force, which of the following correctly describes the situation?



[JEE Main 2025, 23 Jan (Shift 1)]

- (A) $\frac{a}{r} \approx 3$ (B) $\frac{a}{r} \approx 20$
(C) $\frac{a}{r} \approx 0.5$ (D) $\frac{a}{r} \approx 10$

- Q12** Two point charges $-4\mu\text{C}$ and $4\mu\text{C}$, constituting an electric dipole, are placed at $(-9, 0, 0)$ cm and $(9, 0, 0)$ cm in a uniform electric field of strength 10^4 NC^{-1} . The work done on the dipole in rotating it from the equilibrium through 180° is :

[JEE Main 2025, 23 Jan (Shift 2)]

- (A) 12.4 mJ (B) 14.4 mJ
(C) 18.4 mJ (D) 16.4 mJ

- Q13** Two charges $7\mu\text{C}$ and $-4\mu\text{C}$ are placed at $(-7 \text{ cm}, 0, 0)$ and $(7 \text{ cm}, 0, 0)$ respectively. Given, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$, the electrostatic potential energy of the charge configuration is :

[JEE Main 2025, 23 Jan (Shift 2)]

- (A) -2.0 J (B) -1.5 J
(C) -1.2 J (D) -1.8 J

- Q14** An electric charge $10^{-6} \mu\text{C}$ is placed at origin $(0, 0)m$ of $X - Y$ co-ordinate system. Two points P and Q are situated at $(\sqrt{3}, \sqrt{3})m$ and

$(\sqrt{6}, 0)m$ respectively. The potential difference between the points P and Q will be :

[JEE Main 2024, 27 Jan (Shift 1)]

- (A) $\sqrt{6}V$ (B) $3V$
(C) $0V$ (D) $\sqrt{3}V$

- Q15** Two charges of $5Q$ and $-2Q$ are situated at the points $(3a, 0)$ and $(-5a, 0)$ respectively. The electric flux through a sphere of radius $4a$ having center at origin is :

[JEE Main 2024, 29 Jan (Shift 1)]

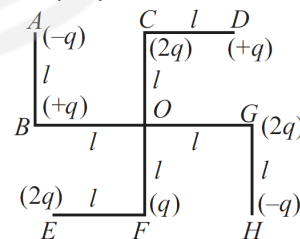
- (A) $\frac{5Q}{\epsilon_0}$
(B) $\frac{2Q}{\epsilon_0}$
(C) $\frac{3Q}{\epsilon_0}$
(D) $\frac{7Q}{\epsilon_0}$

- Q16** An electric field is given by $(6\hat{i} + 5\hat{j} + 3\hat{k}) \text{ N/C}$. The electric flux through a surface area $30\hat{i} \text{ m}^2$ lying in YZ -plane (in SI unit) is :

[JEE Main 2024, 29 Jan (Shift 2)]

- (A) 180 (B) 150
(C) 90 (D) 60

- Q17** What will be the magnitude of electric field at point O as shown in the figure? Each side of the figure is l and perpendicular to each other?



- (A) $\frac{1}{4\pi\epsilon_0} \frac{q}{l^2}$
(B) $\frac{1}{4\pi\epsilon_0} \frac{q}{(2l^2)} (2\sqrt{2} - 1)$
(C) $\frac{q}{4\pi\epsilon_0 (2l^2)}$
(D) $\frac{1}{4\pi\epsilon_0} \frac{2q}{(2l^2)} (\sqrt{2})$

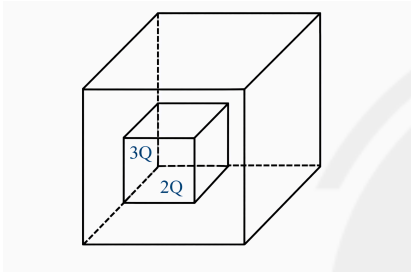


- Q18** Force between two point charges q_1 and q_2 placed in vacuum at ' r ' cm apart is F . Force between them when placed in a medium having dielectric constant $K = 5$ at ' $r/5$ ' cm apart will be:

[JEE Main 2024, 31 Jan (Shift 2)]

- (A) $F/5$
 (B) $25 F$
 (C) $5F$
 (D) $F/25$

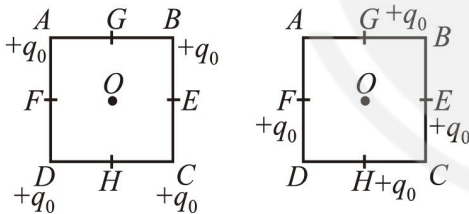
- Q19** C_1 and C_2 are two hollow concentric cubes enclosing charges $2Q$ and $3Q$ respectively as shown in figure. The ratio of electric flux passing through C_1 and C_2 is:



[JEE Main 2024, 01 Feb (Shift 2)]

- (A) 3 : 2
 (B) 5 : 2
 (C) 2 : 5
 (D) 2 : 3

Q20



Configuration (1) Configuration (2)

In the first configuration (1) as shown in the figure, four identical charges (q_0) are kept at the corners A , B , C and D of square of side length ' a '. In the second configuration (2), the same charges are shifted to mid points G , E , H and F , of the square. If $K = \frac{1}{4\pi\epsilon_0}$, the difference between the potential energies of configuration (2) and (1) is given by :

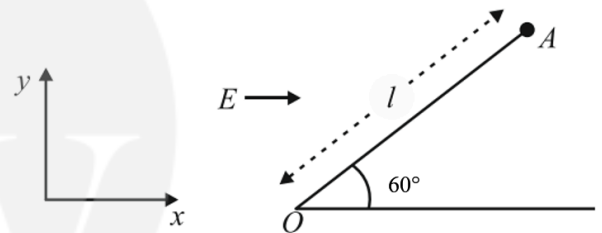
- (A) $\frac{Kq_0^2}{a} (4\sqrt{2} - 2)$ (B) $\frac{Kq_0^2}{a} (3\sqrt{2} - 2)$
 (C) $\frac{Kq_0^2}{a} (4 - 2\sqrt{2})$ (D) $\frac{Kq_0^2}{a} (3 - \sqrt{2})$

- Q21** A small uncharged conducting sphere is placed in contact with an identical sphere but having $4 \times 10^{-8} C$ charge and then removed to a distance such that the force of repulsion between them is $9 \times 10^{-3} N$. The distance between them is (Take $\frac{1}{4\pi\epsilon_0}$ as 9×10^9 in SI units)

[JEE Main 2025, 24 Jan (Shift 2)]

- (A) 2 cm (B) 4 cm
 (C) 1 cm (D) 3 cm

- Q22** A particle of mass ' m ' and charge ' q ' is fastened to one end ' A ' of a massless string having equilibrium length l , whose other end is fixed at point ' O '. The whole system is placed on a frictionless horizontal plane and is initially at rest. If uniform electric field is switched on along the direction as shown in figure, then the speed of the particle when it crosses the X -axis is:



[JEE Main 2025, 28 Jan (Shift 1)]

- (A) $\sqrt{\frac{2qEl}{m}}$ (B) $\sqrt{\frac{qEl}{4m}}$
 (C) $\sqrt{\frac{qEl}{m}}$ (D) $\sqrt{\frac{qEl}{2m}}$

- Q23** Three infinitely long wires with linear charge density λ are placed along the X -axis, Y -axis and Z -axis respectively. Which of the following denotes an equipotential surface?

- (A) $(x^2 + y^2)(y^2 + z^2)(z^2 + x^2) = \text{constant}$
 (B) $xy + yz + zx = \text{constant}$
 (C) $xyz = \text{constant}$
 (D) $(x + y)(y + z)(z + x) = \text{constant}$

- Q24** Two charged conducting spheres of radii a and b are connected to each other by a conducting

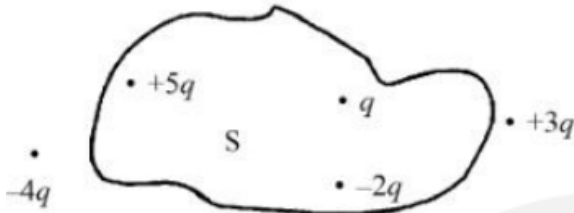


wire. The ratio of charges of the two spheres respectively is:

[JEE Main 2024, 08 Apr (Shift 1)]

- (A) $\sqrt{ab}$ (B) $\frac{b}{a}$
(C) $\frac{a}{b}$ (D) $\frac{a^2}{b^2}$

- Q25** Five charges $+q, +5q, -2q, +3q$ and $-4q$ are situated as shown in the figure. The electric flux due to this configuration through the surface S is :



[JEE Main 2024, 09 Apr (Shift 2)]

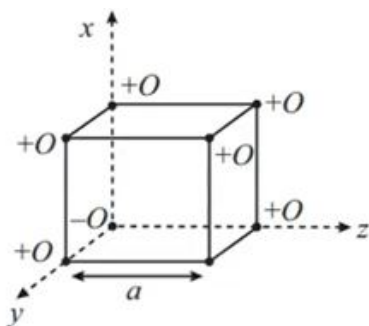
- (A) $\frac{q}{\epsilon_0}$
(B) $\frac{3q}{\epsilon_0}$
(C) $\frac{4q}{\epsilon_0}$
(D) $\frac{5q}{\epsilon_0}$

- Q26** Two identical conducting spheres P and S with charge Q on each, repel each other with a force 16 N . A third identical uncharged conducting sphere R is successively brought in contact with the two spheres. The new force of repulsion between P and S is :

[JEE Main 2024, 06 Apr (Shift 2)]

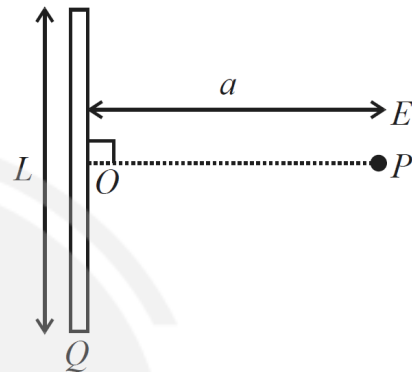
- (A) 1 N (B) 4 N
(C) 12 N (D) 6 N

- Q27** A cube of side a has point charges $+Q$ located at each of its vertices except at the origin where the charge is $-Q$. The electric field at the center of cube is



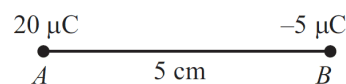
- (A) $\frac{2Q}{3\sqrt{3}\pi\epsilon_0 a^2}(\hat{x} + \hat{y} + \hat{z})$
(B) $\frac{Q}{3\sqrt{3}\pi\epsilon_0 a^2}(\hat{x} + \hat{y} + \hat{z})$
(C) $\frac{-2Q}{3\sqrt{3}\pi\epsilon_0 a^2}(\hat{x} + \hat{y} + \hat{z})$
(D) $\frac{-Q}{3\sqrt{3}\pi\epsilon_0 a^2}(\hat{x} + \hat{y} + \hat{z})$

- Q28** Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q . The distance of the point P from the centre of the rod is $a = \frac{\sqrt{3}}{2}L$.



- (A) $\frac{\sqrt{3}Q}{4\pi\epsilon_0 L^2}$
(B) $\frac{Q}{3\pi\epsilon_0 L^2}$
(C) $\frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$
(D) $\frac{Q}{4\pi\epsilon_0 L^2}$

- Q29** Two particles A and B having charges $20\mu\text{C}$ and $-5\mu\text{C}$ respectively are held with a separation of 5 cm . At what position a third charged particle should be placed so that it does not experience a net electric force?



- (A) At 5 cm from $20\mu\text{C}$ on the left side of system
(B) At 5 cm from $-5\mu\text{C}$ on the right side
(C) At 1.25 cm from $-5\mu\text{C}$ between two charges



(D) At midpoint between two charges

- Q30** Two isolated metallic solid spheres of radii R and $2R$ are charged such that both have same charge density σ . The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ' . The ratio $\frac{\sigma'}{\sigma}$ is: [JEE Main 2023, 30 Jan (Shift 1)]
- (A) $\frac{9}{4}$
 (B) $\frac{4}{3}$
 (C) $\frac{5}{3}$
 (D) $\frac{5}{6}$

- Q31** Two identical charged particles each having a mass 10 g and charge $2 \times 10^{-7} \text{ C}$ are placed on a horizontal table with separation of L between them such that they stay in limited equilibrium. If coefficient of friction between each particle and table is 0.25, find value of L (in cm)

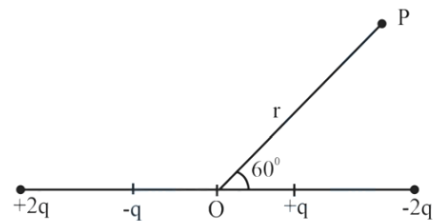
- Q32** An electron is moving under the influence of the electric field of a uniformly charged infinite plane sheet S having surface charge density $+\sigma$. The electron at $t = 0$ is at a distance of 1 m from S and has a speed of 1 m/s. The maximum value of σ if the electron strikes S at $t = 1 \text{ s}$ is $\alpha \left[\frac{m\epsilon_0}{e} \right] \frac{C}{m^2}$ the value of α is

[JEE Main 2024, 29 Jan (Shift 1)]

- Q33** Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 37° with each other. When suspended in a liquid of density 0.7 g/cm^3 , the angle remains same. If density of material of the sphere is 1.4 g/cm^3 , the dielectric constant of the liquid is ----- ($\tan 37^\circ = \frac{3}{4}$)

[JEE Main 2024, 30 Jan (Shift 2)]

- Q34** The distance between charges $+q$ and $-q$ is $2l$ and between $+2q$ and $-2q$ is $4l$. The electrostatic potential at point P at a distance r from centre O is $-\alpha \frac{ql}{r^2} \times 10^9 \text{ V}$, where the value of α is (Use $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$)



[JEE Main 2024, 31 Jan (Shift 2)]

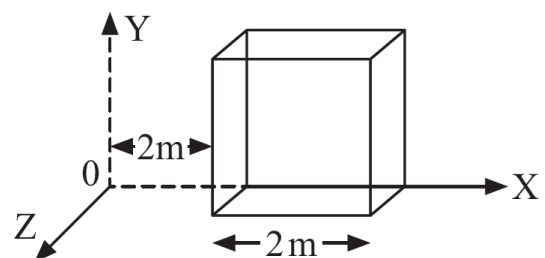
- Q35** Suppose a uniformly charged wall provides a uniform electric field of $2 \times 10^4 \text{ N/C}$ normally. A charged particle of mass 2 g being suspended through a silk thread of length 20 cm and remain stayed at a distance of 10 cm from the wall. Then the charge on the particle will be $\frac{1}{\sqrt{x}} \mu\text{C}$ where $x = \text{_____}$. [use $g = 10 \text{ m/s}^2$]

[JEE Main 2024, 01 Feb (Shift 2)]

- Q36** An infinite plane sheet of charge having uniform surface charge density $+\sigma_s \text{ C/m}^2$ is placed on x - y plane. Another infinitely long line charge having uniform linear charge density $+\lambda_e \text{ C/m}$ is placed at $z = 4 \text{ m}$ plane and parallel to y -axis. If the magnitude values $|\sigma_s| = 2 |\lambda_e|$ then at point $(0, 0, 2)$, the ratio of magnitudes of electric field values due to sheet charge to that of line charge is $\pi \sqrt{n} : 1$. The value of n is _____.

[JEE Main 2024, 04 Apr (Shift 1)]

- Q37** An electric field $\vec{E} = 2x\hat{i} \text{ NC}^{-1}$ exists in space. A cube of side 2 m is placed in the space as per figure given below. The electric flux through the cube is Nm^2/C .



[JEE Main 2024, 09 Apr (Shift 2)]

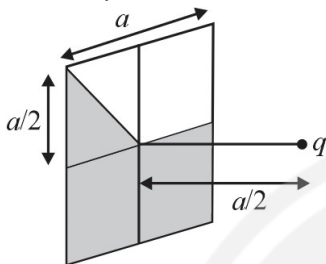
- Q38** A cubical volume is bounded by the surfaces $x = 0, x = a, y = 0, y = a, z = 0, z = a$. The electric field in the region is given by $\vec{E} = E_0 x \hat{i}$. Where $E_0 = 4 \times 10^4 \text{ NC}^{-1} \text{ m}^{-1}$.



If $a = 2 \text{ cm}$, the charge contained in the cubical volume is $Q \times 10^{-14} \text{ C}$. The value of Q is $___$ (Take $\epsilon_0 = 9 \times 10^{-12} \text{ C}^2/\text{Nm}^2$)

[JEE Main 2023, 1 Feb (Shift 2)]

- Q39** A square loop of sides $a = 1 \text{ m}$ is held normally in front of a point charge $q = 1 \text{ C}$. The charge q is placed at a perpendicular distance of $\frac{a}{2}$ from square loop as shown. The flux of the electric field through the shaded region is $\frac{5}{p} \times \frac{1}{\epsilon_0} \frac{\text{Nm}^2}{\text{C}}$, where the value of p is $______$.

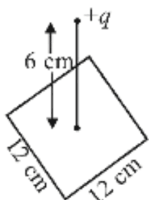


- Q40** The volume charge density of a sphere of radius 6 m is $2\mu\text{Ccm}^{-3}$. The number of lines of force per unit surface area coming out from the surface of the sphere is $______ \times 10^{10} \text{ NC}^{-1}$. [Given: Permittivity of vacuum $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$]

- Q41** The electric potential at the surface of an atomic nucleus ($z = 50$) of radius $9 \times 10^{-13} \text{ cm}$ is $______$

[JEE Main 2024, 27 Jan (Shift 2)]

- Q42** A point charge of $+12\mu\text{C}$ is at a distance 6 cm vertically above the centre of a square of side 12 cm as shown in below figure. The magnitude of the electric flux through the square will be $\times 1000 \text{ Nm}^2/\text{C}$



[JEE Main 2021, 24 Feb (Shift 2)]

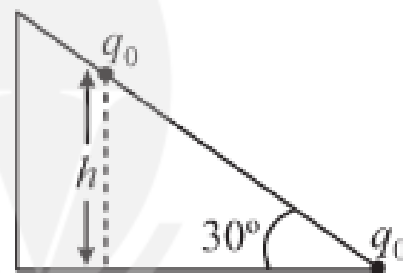
- Q43** 64 identical drops each charged upto potential of 10 mV are combined to form a bigger drop.

The potential of the bigger drop will be $______ \text{ mV}$.
[JEE Main 2023, 12 Apr (Shift 1)]

- Q44** Three concentric spherical metallic shells X , Y and Z of radius a , b and c respectively [$a < b < c$] have surface charge densities σ , $-\sigma$ and σ , respectively. The shells X and Z are at same potential. If the radii of X & Y are 2 cm and 3 cm , respectively. The radius of shell Z is cm .

[JEE Main 2023, 10 Apr (Shift 1)]

- Q45** As shown in the figure, a configuration of two equal point charges ($q_0 = +2\mu\text{C}$) is placed on an inclined plane. Mass of each point charge is 20 g . Assume that there is no friction between charge and plane. For the system of two point charges to be in equilibrium (at rest) the height $h = x \times 10^{-3} \text{ m}$. The value of x is $______ \text{ mm}$. (Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$, $g = 10 \text{ ms}^{-2}$)



[JEE Main 2023, 31 Jan (Shift 1)]



Answer Key

Q1 (B)
Q2 (A)
Q3 (D)
Q4 (D)
Q5 (B)
Q6 (B)
Q7 (D)
Q8 (A)
Q9 (D)
Q10 (B)
Q11 (A)
Q12 (B)
Q13 (D)
Q14 (C)
Q15 (A)
Q16 (A)
Q17 (B)
Q18 (C)
Q19 (C)
Q20 (B)
Q21 (A)
Q22 (C)
Q23 (A)

Q24 (C)
Q25 (C)
Q26 (D)
Q27 (C)
Q28 (C)
Q29 (B)
Q30 (D)
Q31 12
Q32 8
Q33 2
Q34 27
Q35 3
Q36 16
Q37 16
Q38 288
Q39 48
Q40 45
Q41 8
Q42 (226)
Q43 160
Q44 5
Q45 300



[Master NCERT with PW Books APP](#)